Modeling Transformations

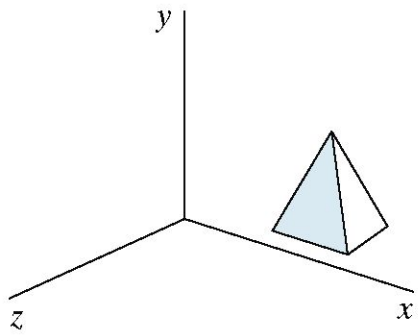
2D TRANSFORMATIONS

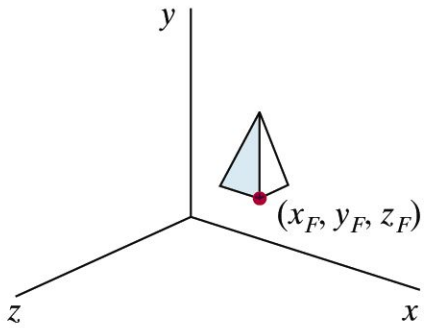
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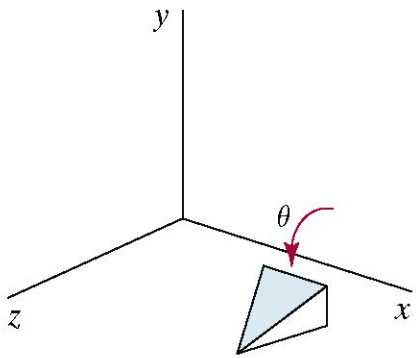
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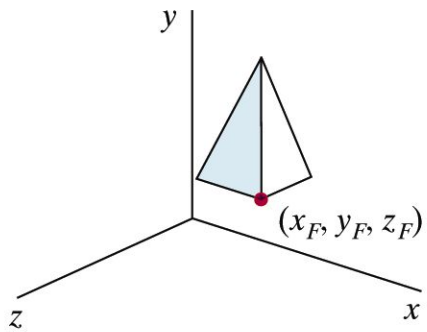
Why Transformations?

- In graphics, once we have an object described, transformations are used to move that object, scale it and rotate it





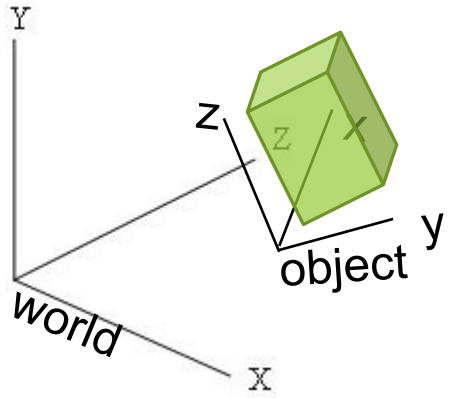


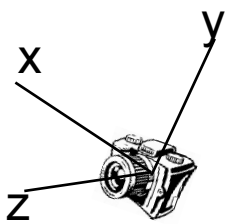


Images taken from Hearn & Baker, "Computer Graphics with OpenGL" (2004)

Use of Transformations

- Modeling
  - Build complex models by positioning simple components
  - Transform from object coordinates to world coordinates
- Viewing
  - Placing the virtual camera in the world
  - Specifying transformation from world to camera coordinates
- Animation
  - Vary transformations over time to create motion





Modeling Transformations

- ~ places the objects defined by their own coordinate system into a common scene of world coordinate system.
- Moving or transforming an object can be achieved by
  - **Geometric transformation** – changes in orientation, size and shape of objects are accomplished by altering coordinate description of objects.
  - **Coordinate Transformation** – same effect is attained by holding the object stationary while coordinate system is transformed relative to object.

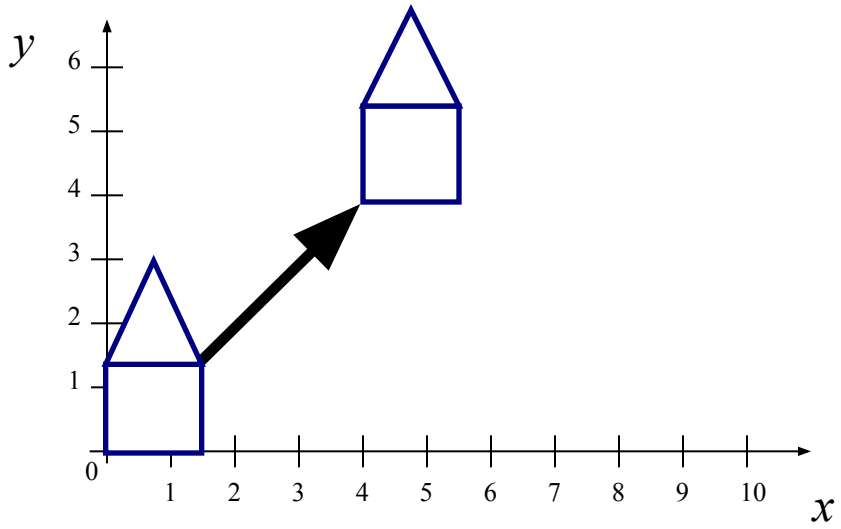
Modeling Transformations

- We will go with Geometric transformation
  - Basic Transformations
    - Translation
    - Rotation
    - Scaling
  - Some else are
    - Shear
    - Mirror reflection
  - Homogeneous coordinate system
  - Composition of transformations

Translation

- Simply moves an object from one position to another
  - Object is displaced a given distance and direction

$$x_{new} = x_{old} + dx \qquad y_{new} = y_{old} + dy$$



Note: House shifts position relative to origin

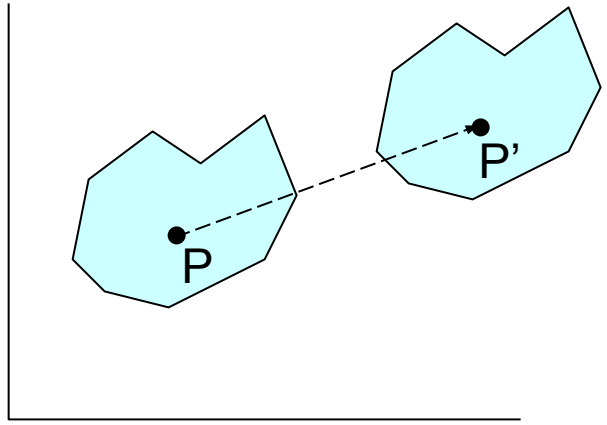
Translation – 2D

- If displacement is the vector $V = t_x I + t_y J$, new point P′(x′,y′) can be found by adding translation distance (t_x,t_y) to the old point P(x,y)

$$i.e. \quad x' = x + t_x \;, \quad y' = y + t_y \quad \text{--- (i)}$$

Representing the points and displacement as a column vector

$$P = \begin{bmatrix} x \\ y \end{bmatrix} \quad P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \quad T = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

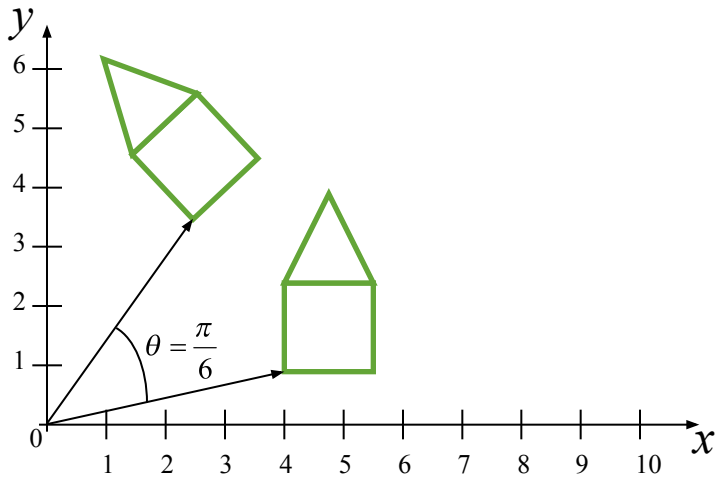


Eq. (i) becomes

$$P' = P + T$$

Rotation – 2D

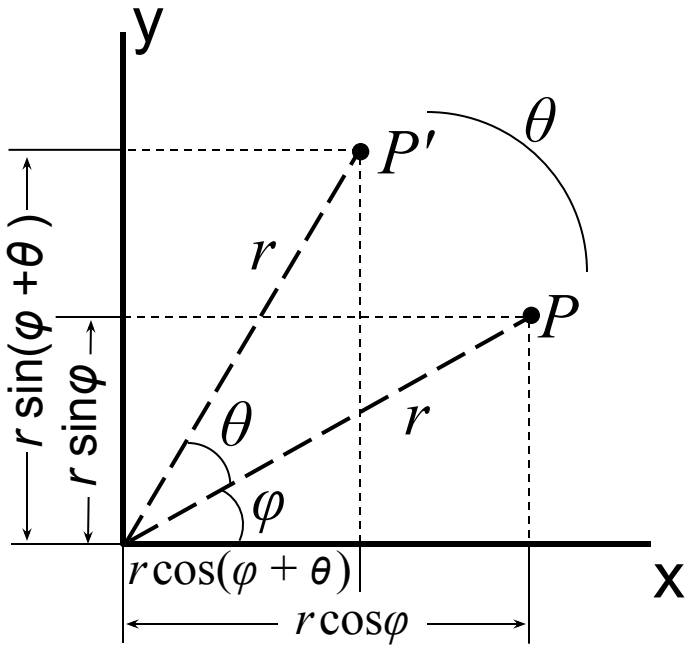
- In 2d rotation, object is repositioned along a circular path in xy plane.
  - Rotates all coordinates by a specified angle
- For rotation we specify
  - Rotation angle (θ)
    - Counter clockwise – positive angle
    - Clockwise – negative angle
  - Pivot point (x_r,y_r) – center of rotation
- What is the transformation eqn?



Rotation – 2D, origin as pivot point

$$P = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r\cos\phi \\ r\sin\phi \end{bmatrix}$$

$$P' = \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} r\cos(\phi+\theta) \\ r\sin(\phi+\theta) \end{bmatrix}$$

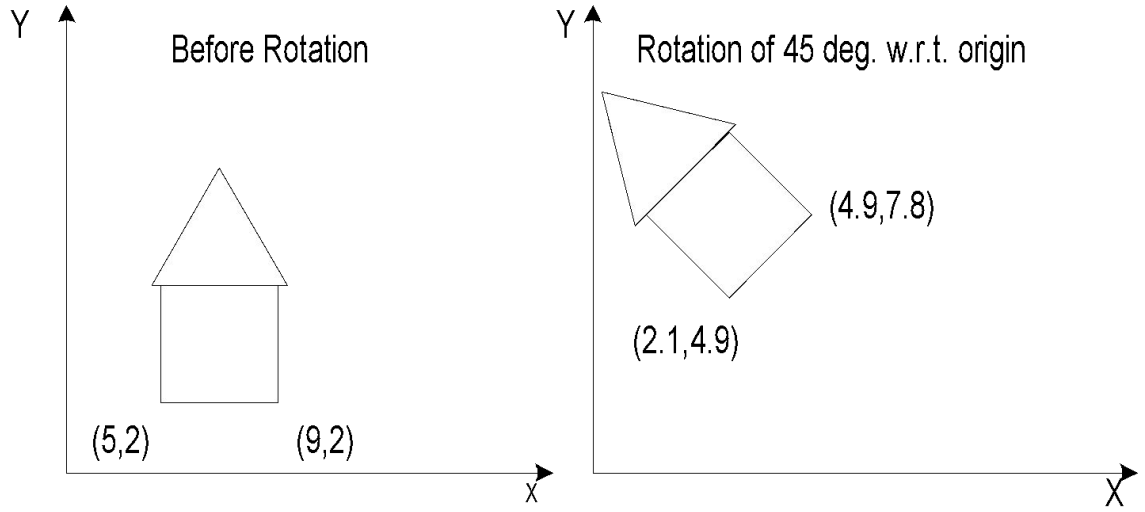


$$\text{expand}\,(\phi+\theta) \Rightarrow \begin{cases} x' = r\cos\phi\cos\theta - r\sin\phi\sin\theta \\ y' = r\cos\phi\sin\theta + r\sin\phi\cos\theta \end{cases}$$

$$\text{but} \quad \begin{matrix} x = r\cos\phi \\ y = r\sin\phi \end{matrix} \Rightarrow \begin{matrix} x' = x\cos\theta - y\sin\theta \\ y' = x\sin\theta + y\cos\theta \end{matrix}$$

Rotation – 2D

$$x' = x \cdot \cos\theta - y \cdot \sin\theta$$
$$y' = x \cdot \sin\theta + y \cdot \cos\theta$$



This eq can be represented in matrix form as -

$$P' = R \cdot P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \cdot \cos\theta - y \cdot \sin\theta \\ x \cdot \sin\theta + y \cdot \cos\theta \end{bmatrix} = \begin{bmatrix} \quad \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

Scaling – 2D

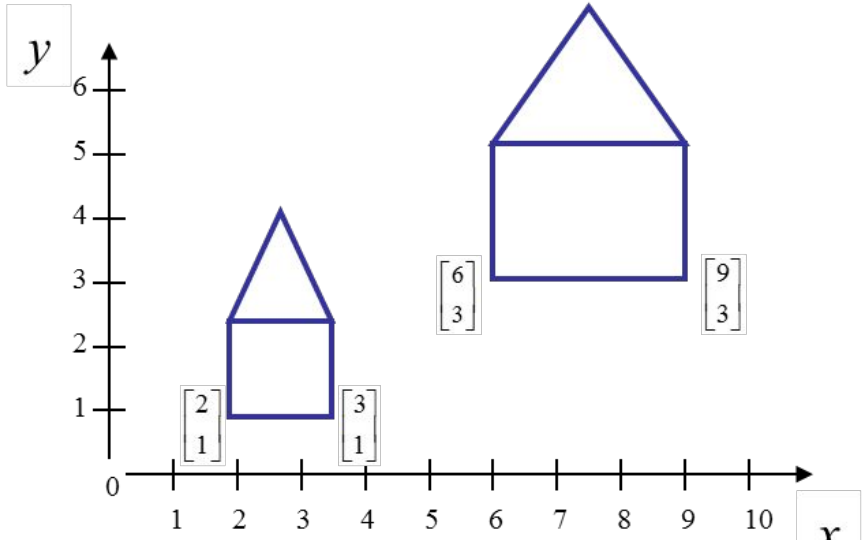
- Scaling alters the size of an object
- Transformed coordinates P′(x′, y′) are found by multiplying the coordinate values P(x, y) of each vertex of object by **scaling factor** s_x and s_y.
- ie.

$$x' = s_x \cdot x \quad , \quad y' = s_y \cdot y$$

- In matrix form

$$P' = S \cdot P$$

$$\begin{bmatrix} x \cdot s_x \\ y \cdot s_y \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

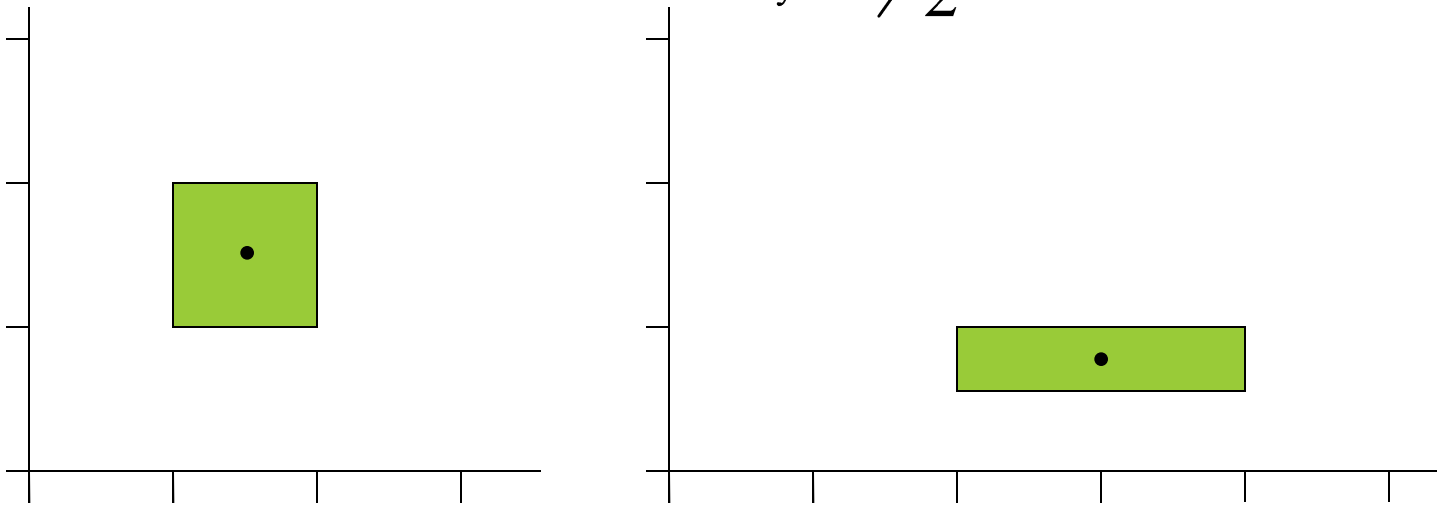


WATCH OUT: Objects grow and move!

Scaling – 2D

- Types of Scaling:
  - Differential (s_x != s_y)
  - Uniform ($s_x = s_y$)
- After scaling, new object is located at a different position relative to origin

$$scale\ factors \quad s_x = 2,\ s_y = \frac{1}{2}$$



Homogeneous Coordinates

- A point (x, y) can be re-written in **homogeneous coordinates** as (x_h, y_h, h)
- The **homogeneous parameter** h is a non-zero value such that:

$$x = \frac{x_h}{h} \qquad y = \frac{y_h}{h}$$

- We can then write any point (x, y) as (h_x, h_y, h)
- We can conveniently choose $h = 1$ so that (x, y) becomes $(x, y, 1)$

Why Homogeneous Coordinates?

- Mathematicians commonly use homogeneous coordinates as they allow scaling factors to be removed from equations
- Translation, scaling and rotation are expressed (non-homogeneously) as:
  - translation: $P' = P + T$
  - Scale: $P' = S \cdot P$
  - Rotate: $P' = R \cdot P$
- Composition is difficult to express, since translation not expressed as a matrix multiplication
- Using homogeneous coordinates allows us use matrix multiplication to calculate transformations – extremely efficient!

Homogeneous Co-ordinates

- Translation, scaling and rotation are expressed (non-homogeneously) as:
  - translation: P′ = P + T
  - Scale: P′ = S · P
  - Rotate: P′ = R · P
- Composition is difficult to express, since translation not expressed as a matrix multiplication
- Homogeneous coordinates allow all three to be expressed homogeneously, using multiplication of 3×3 matrices
- Each Cartesian position (x,y) is represented by a triple (x_h, y_h, h), where,

$$x = \frac{x_h}{h} \qquad y = \frac{y_h}{h}$$

Homogeneous Co-ordinates

- h can have any non zero value
- So an infinite number of points correspond to (x,y)
- Better to use h = 1
- Point representation in ~ system allows all transformation eq as matrix multiplication, and
- Transformation matrices are
  - For translation

$$x' = x + t_x \quad , \quad y' = y + t_y$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$T_{(t_x, t_y)}$$

Homogeneous Translation

$$x' = x + t_x \quad , \quad y' = y + t_y$$

- Representing the point as a homogeneous column vector
  - The translation of a point by (t_x, t_y) can be written in matrix form as:

$$\mathbf{x}' = \mathbf{T}_{(t_x, t_y)} \quad \mathbf{x}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Translation

- The translation of a point by $(tx,\ ty)$ can be written in matrix form as:

$$T_{(t_x,t_y)} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix}$$

- Representing the point as a homogeneous column vector we perform the calculation as:

$$\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1*x+0*y+tx*1 \\ 0*x+1*y+ty*1 \\ 0*x+0*y+1*1 \end{bmatrix} = \begin{bmatrix} x+tx \\ y+ty \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

i.e. $x' = x + t_x \ , \ y' = y + t_y$

Homogeneous Co-ordinates

- For rotation

$$x' = x \cdot \cos\theta - y \cdot \sin\theta \quad , \quad y' = x \cdot \sin\theta + y \cdot \cos\theta$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$R_{(\theta)}$$

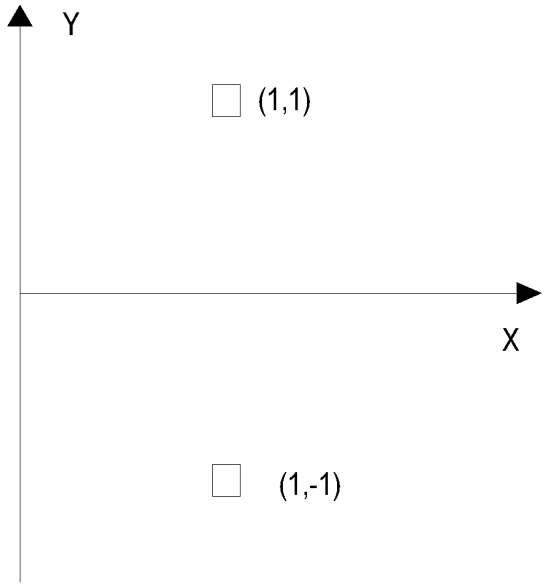
- For Scaling

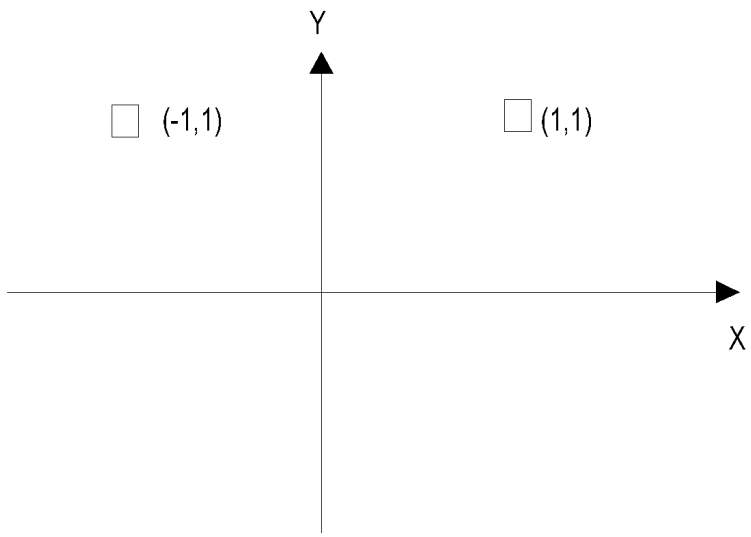
$$x' = s_x \cdot x \quad , \quad y' = s_y \cdot y$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$S_{(s_x, s_y)}$$

Mirror Reflection





Reflection about X - axis

$$x' = x \quad y' = -y$$

$$M_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Y - axis

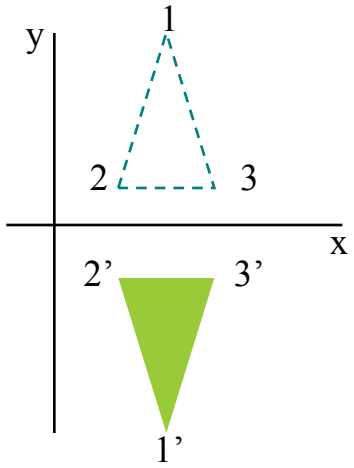
$$x' = -x \quad y' = y$$

$$M_y = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection

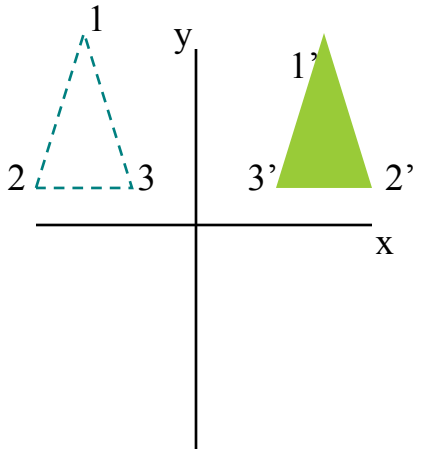
- Reflection with respect to the axis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



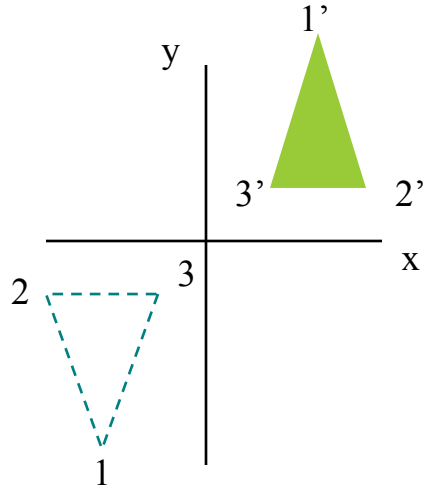
About x axis

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



About y axis

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



About origin

Shearing Transformation

$$x' = x + shx$$
$$y' = y$$

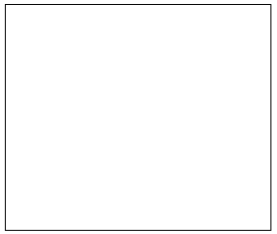
$$x' = x$$
$$y' = shy + y$$

$$x' = x + shx$$
$$y' = shy + y$$

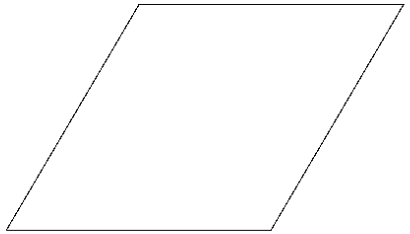
$$SH_{(sh_x)} = \begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$SH_{(sh_y)} = \begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

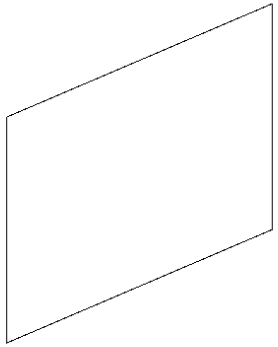
$$SH_{(sh_x,sh_y)} = \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



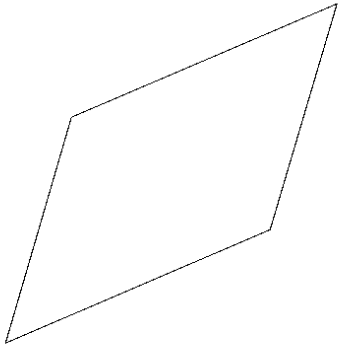
unit cube



Sheared in X direction



Sheared in Y direction



Sheared in both X and Y direction

Inverse 2D - Transformations

- Transformations can easily be reversed using inverse transformations

Translaiton: $T^{-1}_{(tx,ty)} = T_{(-tx,-ty)} = \begin{bmatrix} 1 & 0 & -tx \\ 0 & 1 & -ty \\ 0 & 0 & 1 \end{bmatrix}$

Rotation : $R^{-1}_{(\theta)} = R_{(-\theta)} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Sclaing : $S^{-1}_{(sx,sy)} = S_{(1/sx,\,1/sy)} = \begin{bmatrix} 1/sx & 0 & 0 \\ 0 & 1/sy & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Mirror Ref: $M_x^{-1} = M_x$

$M_y^{-1} = M_y$

$$\mathbf{SH}^{-1}_{(sh_x,sh_y)} = \frac{-1}{1 - sh_x sh_y} \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix}$$

$$cofactor(T) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -tx & -ty & 1 \end{bmatrix}$$

$$adj(T) = \begin{bmatrix} 1 & 0 & -tx \\ 0 & 1 & -ty \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det(T) = 1$$

$$T^{-1} = \frac{1}{\det(T)} adj(T)$$

Composite Transformation

- **Transformations can be combined by matrix multiplication**

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \left(\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$p' = \mathrm{T}(tx, ty)\ \mathrm{R}(\theta)\ \mathrm{S}(sx, sy)\ p$$

- **Efficiency with premultiplication**
  - Matrix multiplication is associative

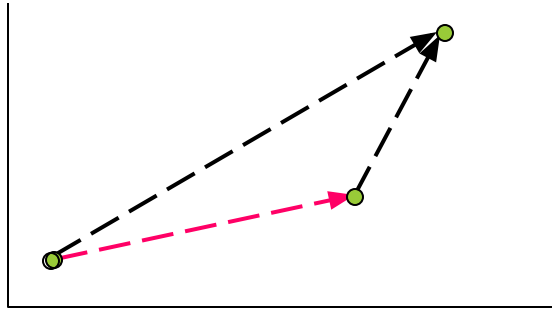
$$p' = (T \times (R \times (S \times p))) \Rightarrow p' = (T \times R \times S) \times p$$

Composite Transformation

- Translation: 2 successive ~ are applied to point P

$$P' = T(t_{x2}, t_{y2}) \cdot [T(t_{x1}, t_{y1}) \cdot P] = [T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1})] \cdot P$$

$$Now,\ T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1}) = \begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix} = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$$

- ie. Successive translations are additive

Composite Transformation - DIY

- Successive Rotation are additive, and
- Successive Scaling are multiplicative

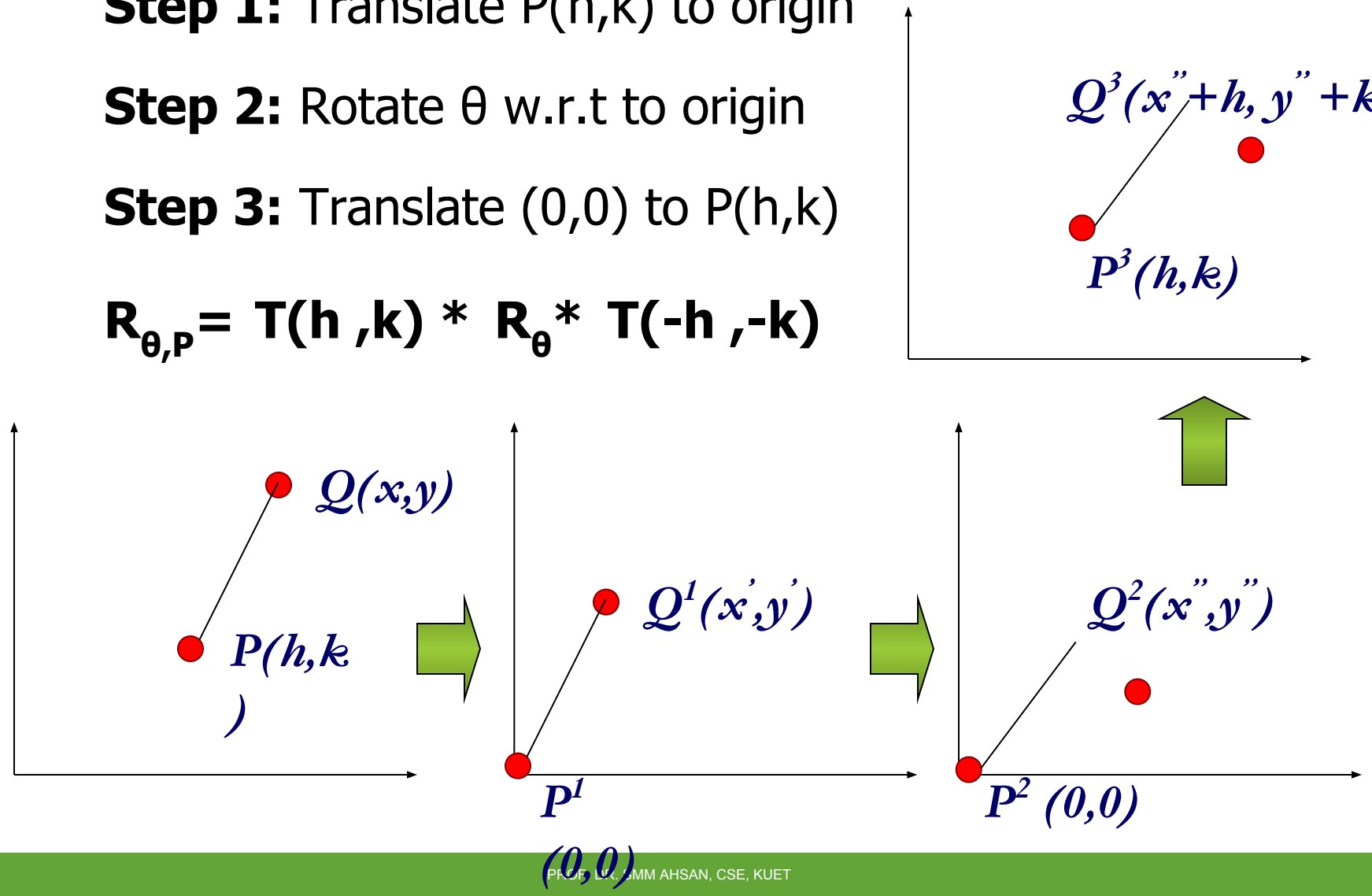
General Pivot point, P(h,k), Rotation of θ : $R_{\theta,P}$

Step 1: Translate P(h,k) to origin

Step 2: Rotate θ w.r.t to origin

Step 3: Translate (0,0) to P(h,k)

$$R_{\theta,P} = T(h,k) * R_{\theta} * T(-h,-k)$$



General Pivot point, P(h,k), Rotation of θ : $R_{\theta,P}$

$$R_{\theta,p} = T_{(h,k)} \bullet R_{\theta} \bullet T_{(-h,-k)}$$

$$R_{\theta,p} = \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & h \\ \sin\theta & \cos\theta & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & h(1-\cos\theta)+k\sin\theta \\ \sin\theta & \cos\theta & k(1-\cos\theta)-h\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

Example

- Rotate the point P(4,5) 45° about the point A(1,1)

$$P' = R_{45^\circ,A} \bullet P = T_{(1,1)} \bullet R_{45^\circ} \bullet T_{(-1,-1)} \bullet P$$

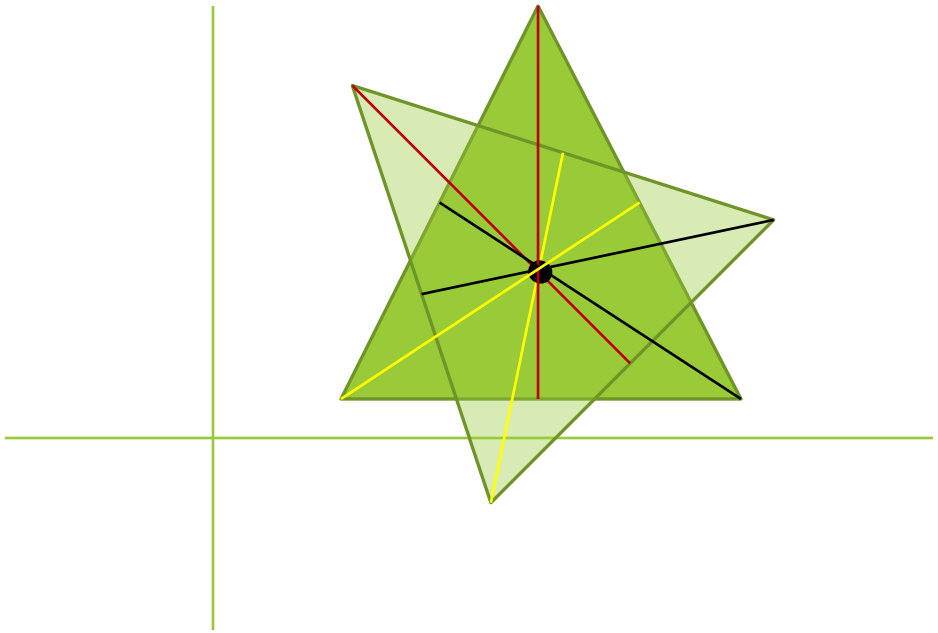
$$= \begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -h \\ 0 & 1 & -k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & h(1-\cos\theta)+k\sin\theta \\ \sin\theta & \cos\theta & k(1-\cos\theta)-h\sin\theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 1(1-1/\sqrt{2})+1.1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1(1-1/\sqrt{2})-1.1/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 1 \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1/\sqrt{2} \\ 1+7/\sqrt{2} \\ 1 \end{bmatrix}$$

DIY

- Rotate the triangle keeping the center fixed



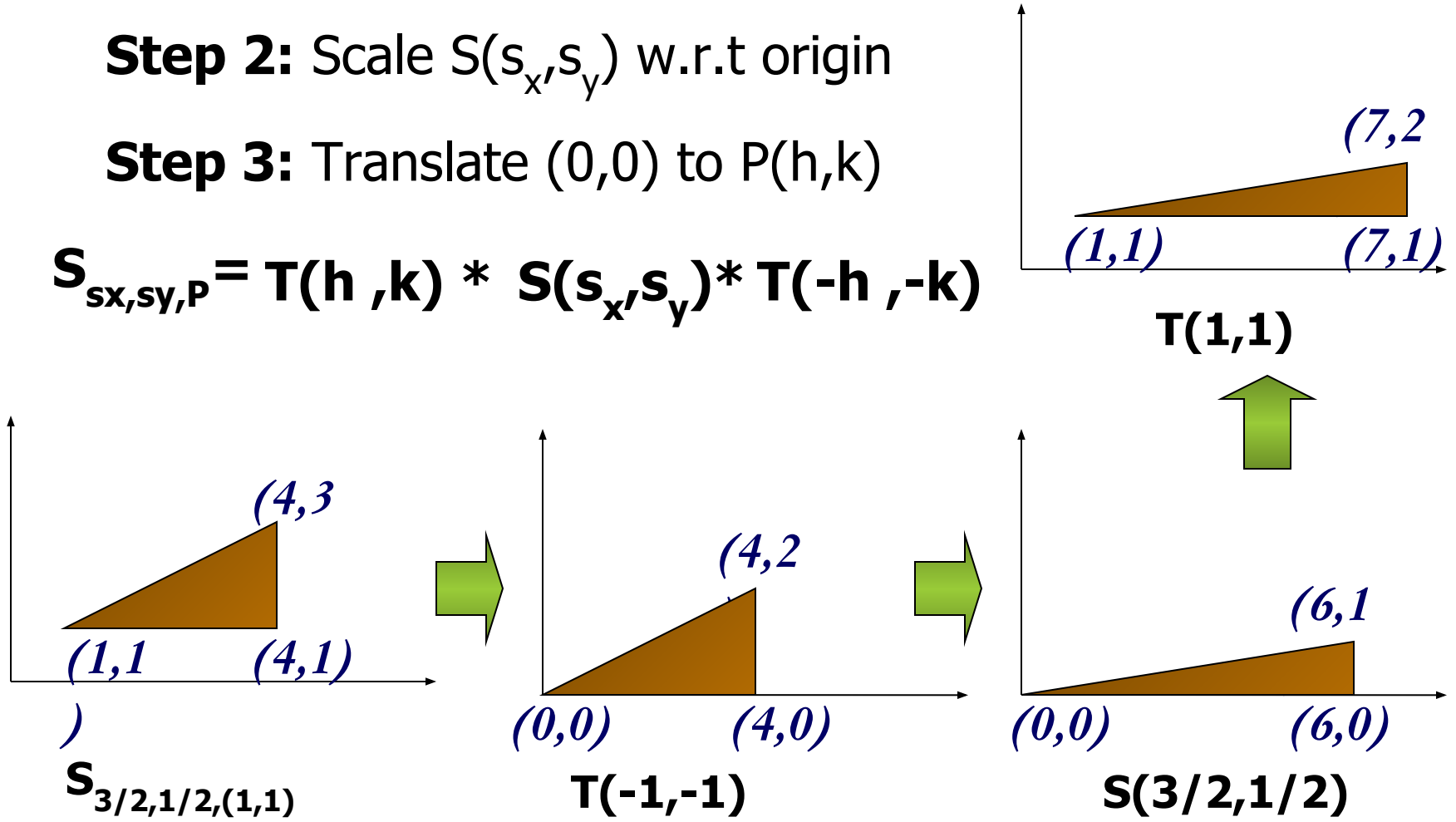
Fixed point P(h,k) Scaling : $S_{sx,sy,p}$

Step 1: Translate P(h,k) to origin

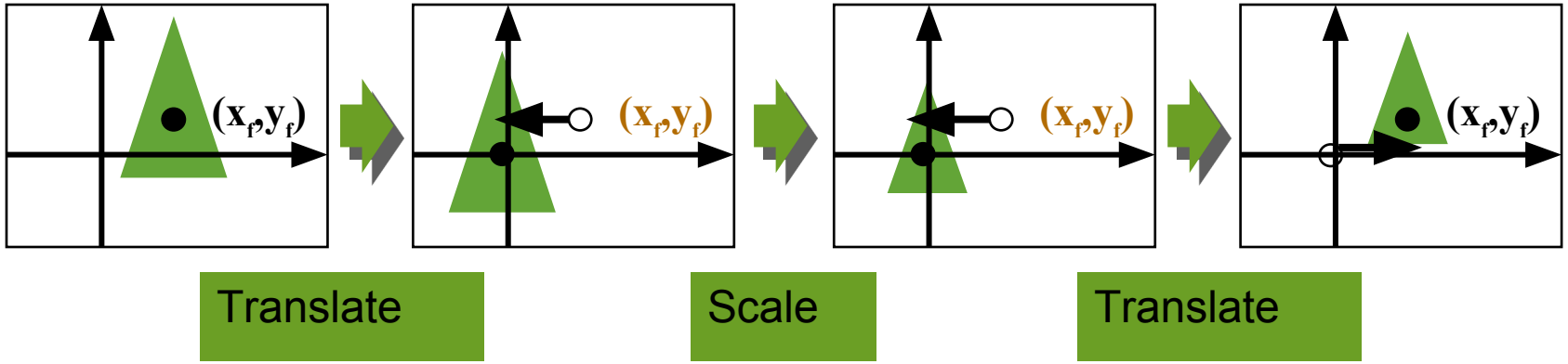
Step 2: Scale S(s_x,s_y) w.r.t origin

Step 3: Translate (0,0) to P(h,k)

$$S_{sx,sy,P} = T(h,k) * S(s_x,s_y) * T(-h,-k)$$



Fixed point P(h,k) Scaling : $S_{sx,sy,p}$



$$T(x_f, y_f) \cdot S(s_x, s_y) \cdot T(-x_f, -y_f) = S(x_f, y_f, s_x, s_y)$$

$$\begin{bmatrix} 1 & 0 & x_f \\ 0 & 1 & y_f \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_f \\ 0 & 1 & -y_f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & x_f(1-s_x) \\ 0 & s_y & y_f(1-s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about line L, M_L

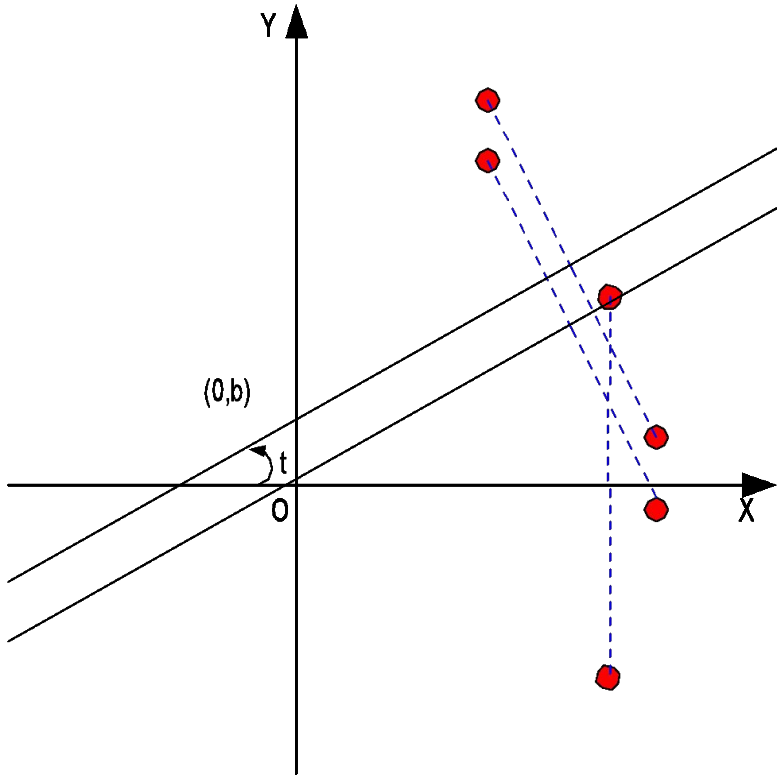
Step 1: Translate (0,b) to origin

Step 2: Rotate -θ degrees

Step 3: Mirror reflect about X-axis

Step 4: Rotate θ degrees

Step 5: Translate origin to (0,b)



$$M_L = T(0, b) * R(\theta) * M_x * R(-\theta) * T(0, -b)$$

Reflection about line L, M_L

$$M_L = T(0, b) * R(\theta) * M_x * R(-\theta) * T(0, -b)$$

$$M_L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & -\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} cos^2\theta - sin^2\theta & 2\sin\theta cos\theta & 0 \\ 2\sin\theta cos\theta & -(cos^2\theta - sin^2\theta) & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{1 + tan^2\theta} \begin{bmatrix} 1 - tan^2\theta & 2tan\theta & 0 \\ 2tan\theta & -(1 - tan^2\theta) & b(1 + tan^2\theta) \\ 0 & 0 & 1 + tan^2\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_L = \frac{1}{m^2 + 1} \begin{bmatrix} 1 - m^2 & 2m & -2bm \\ 2m & m^2 - 1 & 2b \\ 0 & 0 & m^2 + 1 \end{bmatrix}$$

$$\sin(2\theta) = 2\sin\theta\cos\theta = (\sin\theta + \cos\theta)^2 - 1 = \frac{2\tan\theta}{1 + \tan^2\theta}$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \frac{1 - \tan^2\theta}{1 + \tan^2\theta}$$

Problem

Show that two successive reflections about any line (two lines) in the xy plane that intersects the coordinate origin is equivalent to a rotation in the xy plane about the origin

$$M_{L1} = \begin{bmatrix} cos^2\theta_1 - sin^2\theta_1 & 2\sin\theta_1 cos\,\theta_1 & 0 \\ 2\sin\theta_1 cos\,\theta_1 & -(cos^2\theta_1 - sin^2\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta_1 & \sin 2\theta \\ \sin 2\theta_1 & -\cos 2\theta_1 \end{bmatrix} = \begin{bmatrix} \cos A & \sin A \\ \sin A & -\cos A \end{bmatrix}$$

$$M_{L2} = \begin{bmatrix} \cos B & \sin B \\ \sin B & -\cos B \end{bmatrix}$$

$$M_{L2}M_{L1} = \begin{bmatrix} \cos A\cos B + \sin A\sin B & \sin A\cos B - \cos A\sin B \\ \cos A\sin B - \sin A\cos B & \cos A\cos B + \sin A\sin B \end{bmatrix} = \begin{bmatrix} \cos(A-B) & \sin(A-B) \\ -\sin(A-B) & \cos(A-B) \end{bmatrix}$$

Problems to be solved:

Schaum's outline series:

Problems:

- 4.2
- 4.3, 4.4, 4.5 => $R_{\theta,P}$
- 4.6, 4.7, 4.8 => $S_{sx,sy,,P}$
- 4.9, 4.10, 4.11, 4.21 => M_L
- 4.12 => Shearing
- 4.29, 4.34